Université Cheikh Anta Diop

MAPME maps for planning, monitoring & evaluation

SOURCE Soutenabilité et résilience

UVSQ université PARIS-SACLAY

IRD Institut de Recherche pour le Développement

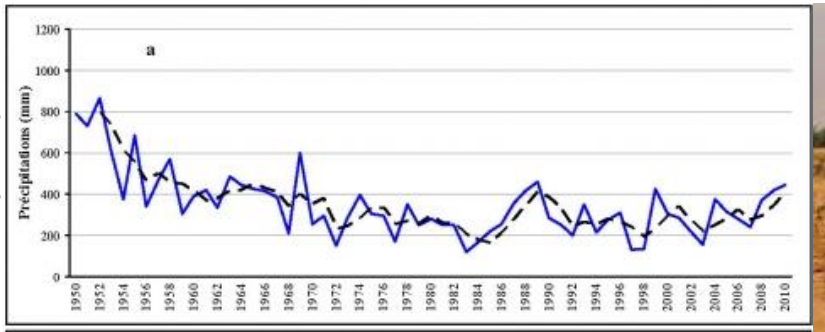
Mapping the State of Land Degradation in a Context of Climate Change at the Scale of Geographical Areas (GGW)

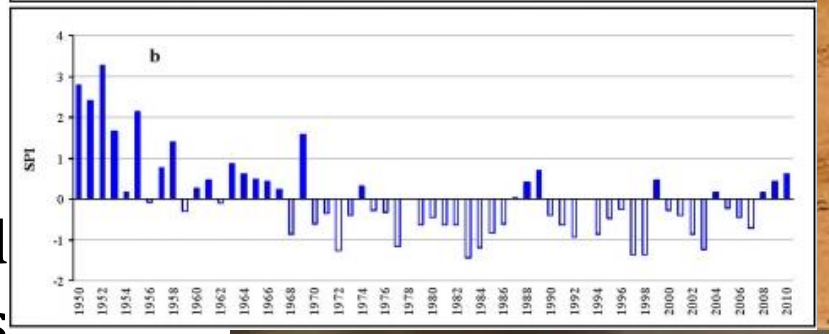
Birane CISSE, Libasse H. SARR, Amadou BARRY, Sidia D. BADIANE

CONTEXT AND OBJECTIVE

Environmental Challenges

Low rainfall patterns exacerbate ecosystem vulnerability. Severe degradation affects plant communities across the region.







Climate Vulnerability

Increasing temperatures and unpredictable precipitation patterns intensify degradation processes across the study area.



Fire Impact

Bush fires contribute significantly to land degradation. These events compromise soil quality and vegetation recovery.

CONTEXT AND OBJECTIVE

1

Baseline Assessment

Establish reference conditions for land degradation levels, document current ecosystem status across target zones.

2

Resilience Building

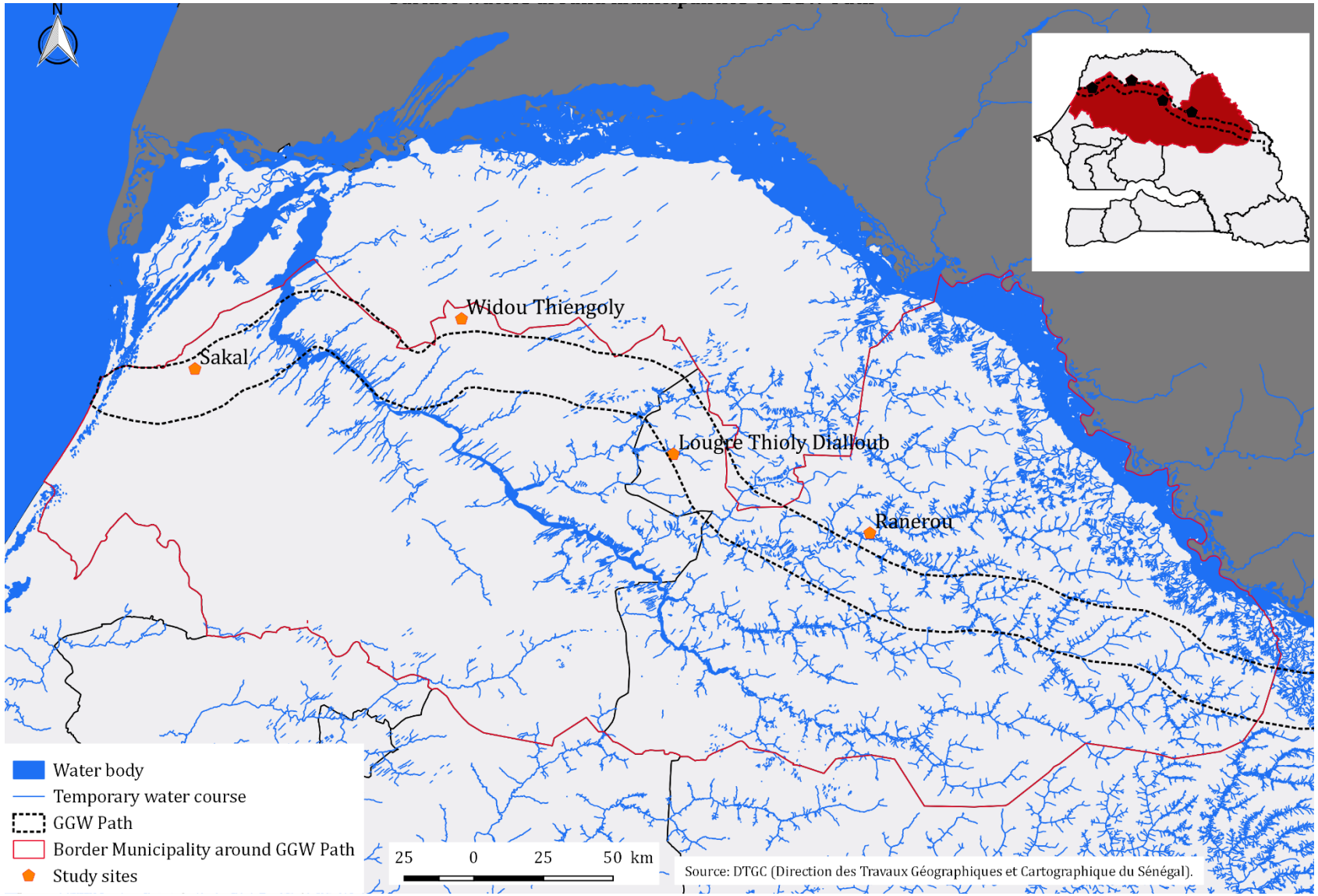
Strengthen ecosystem adaptation capacity, reduce community vulnerability to climate impacts.

3

Strategic Implementation

Support Great Green Wall intervention strategies, provide scientific foundation for restoration efforts.

GEOGRAPHICAL STUDY AREA



- **3 regions** (Louga, Matam and Tambacounda)
- **6 departments** (Bakel, Kanel, Kébémer, Louga, Linguère and Matam)
- **18 municipalities** (Diokoul Diawrigne, Keur Momar Sarr, Labgar, Léona, Mboula, Bélé and Gabou, etc.)

METHODOLOGICAL APPROACH

1

Satellite Data Acquisition

Landsat imagery from 2000 (L5) and 2023 (L8 OLI).
Multi-temporal analysis for degradation assessment.

2

Spectral Index Calculation

NDVI, TCT-B, and BSI indices computed.
Quantitative measures of vegetation and soil conditions.

3

Ground Truth Validation

Field verification of classified areas.
Accuracy assessment through direct observation.

Indices	Formules et atouts	Auteur
NDVI	$(\rho_PIR-\rho_R)/(\rho_PIR+\rho_R)$	Rouse et al. (197
Indice de végétation de différence normalisée		
	Sensible à la vigueur et à la quantité de la végétation	
	Efficacité à discriminer l'état de santé des végétaux	
DVI	$(\rho PIR - \rho R)$	Richardson & W (1977)
Indice de différence de végétation		
MSAVI2	$2\ \rho PIR+1-((2\ \rho PIR+1)2-8(\rho PIR-\rho R))0.5/\ 2$	Qi et al. (1994a)
Indice de végétation ajusté en fonction du sol	Permet de caractériser la végétation en tenant compte de l'humidité du sol.	
TCT-B	$(0.3029*\rho B)+(0.2786*\rho V)+(0.4733*\rho R)+(0.5599*\rho PIR)+(0.508*\rho MIR1)+(0.1872*\rho MIR2)$	Kauth & Thomas
Indice de brillance de la Transfor-mation Tasseled Cap		
	Détection des modifications associées à la végétation, au sol et autres fonctions résultant d'une intervention humaine.	
NDBI	$(\rho_MIR1-\rho_PIR)/(\rho_MIR1+\rho_PIR)$	Zha et al. (2003)
Indice d'urbanisation par diffé-rence normalisée		
BSI	$(\rho R+\rho MIR1)-(\rho PIR+\rho B)(\rho R+\rho MIR1)+(\rho PIR+\rho B)$	
Indice de sol nu		

METHODOLOGICAL APPROCH

Spectral Analysis Framework

Index Combination

RGB system using **NDVI, TCT-B, BSI**.

Optimal spectral combination from 30 tested configurations.

Synthetic Index Development

Average-based composite index calculation.

Five-class degradation intensity classification system.

Intensité de dégradation	*NDVI*	*TCT-B*	*BSI*
Très faible	> 0,50	< 0,36	< -0,03
Faible	0,37 à 0,50	0,36 à 0,40	-0,03 à -0,01
Modérée	0,25 à 0,37	0,40 à 0,44	-0,01 à 0,08
Forte	0,13 à 0,25	0,44 à 0,52	0,08 à 0,13
Très forte	< 0,13	> 0,52	> 0,13

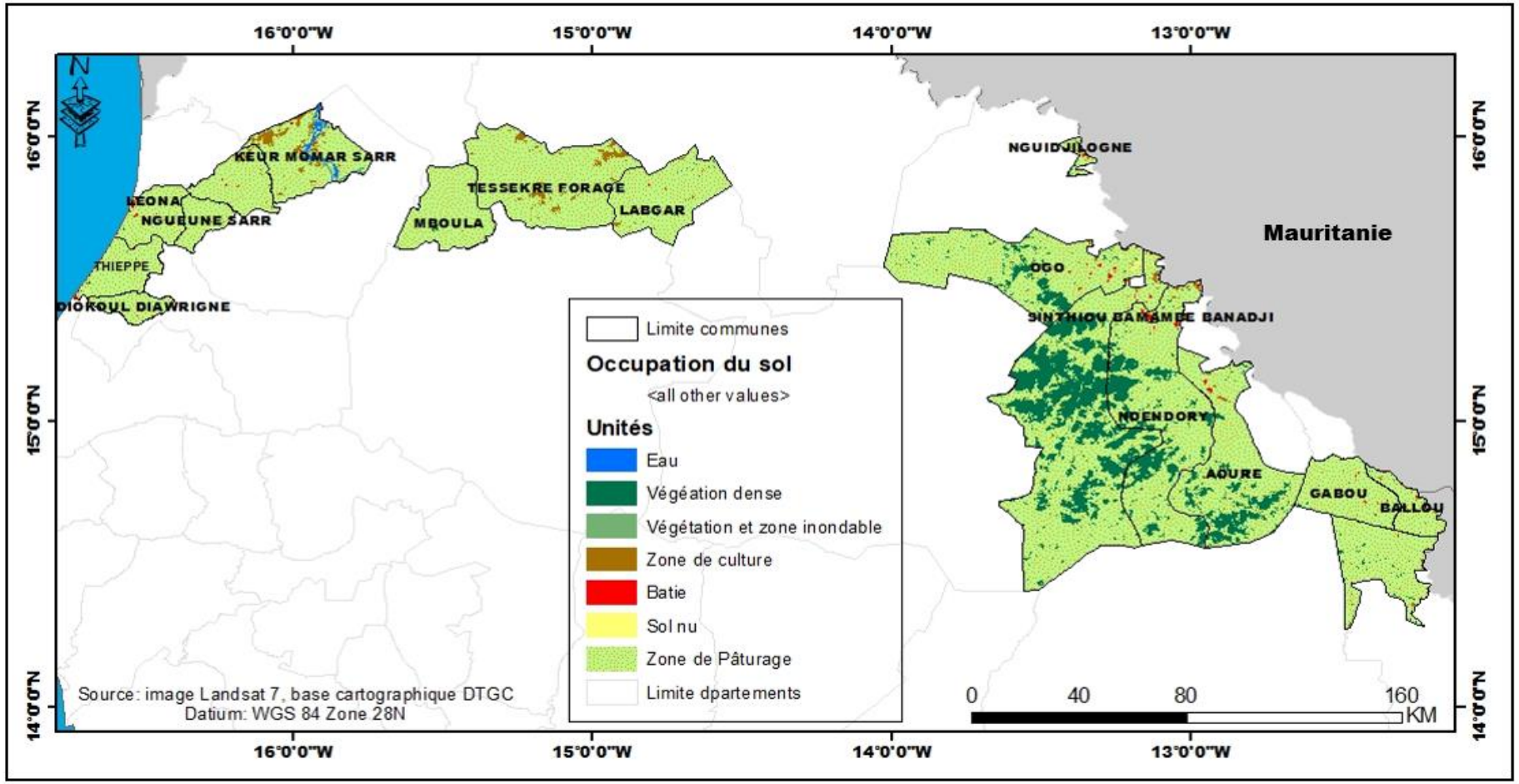
Image Classification

Automated land degradation mapping protocol.

Statistical validation through field reconnaissance campaigns.

RESULTS

Analysis of land use in the target municipalities reveals a clear predominance of open spaces or grazing areas, which cover 85% of the total surface area.



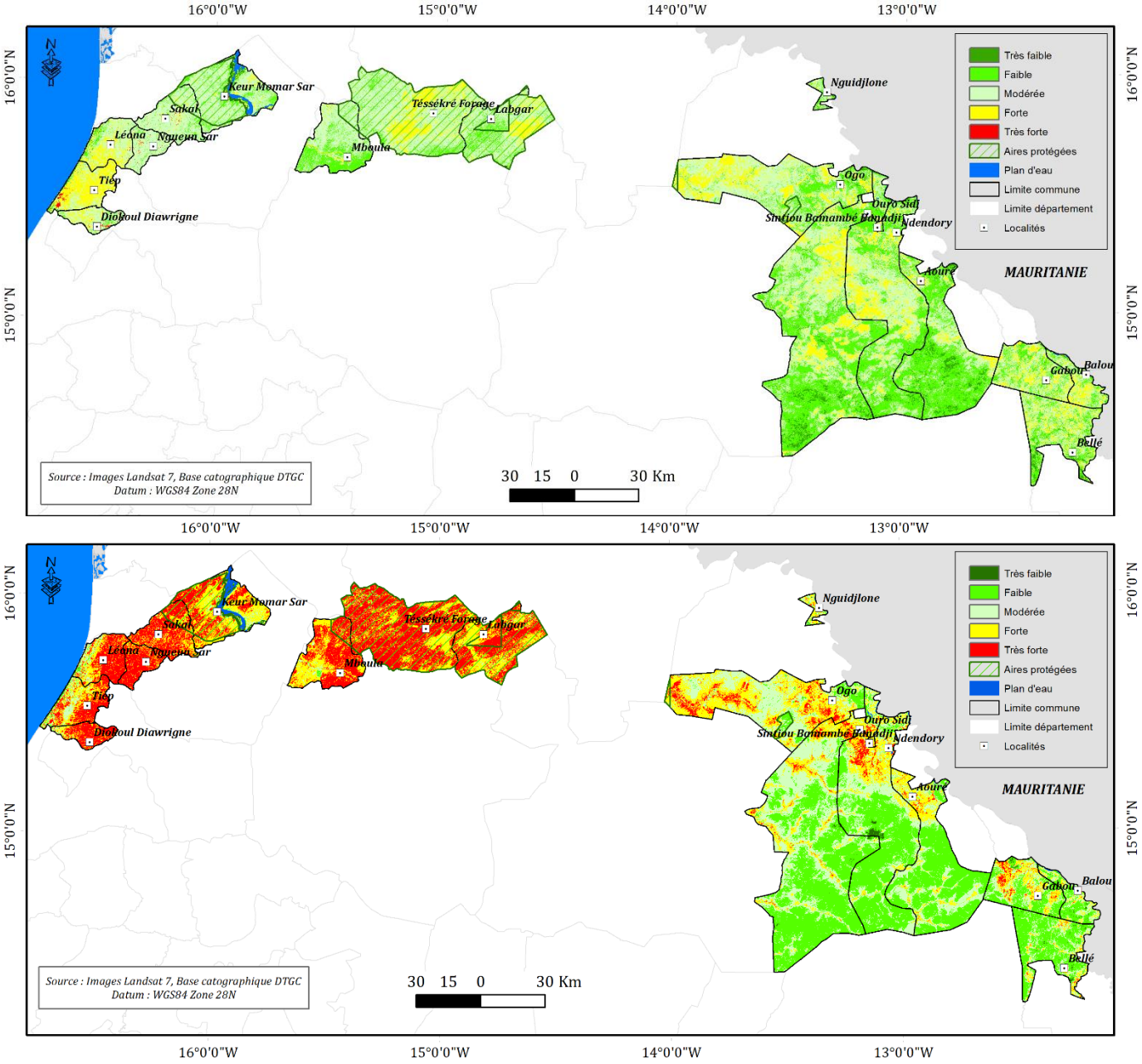
Grazing areas 85%

Vegetation 12.5%

Cultivated areas 1.4%

Other units 0.9% (bare soil, flooded vegetation, water bodies, and built-up areas)

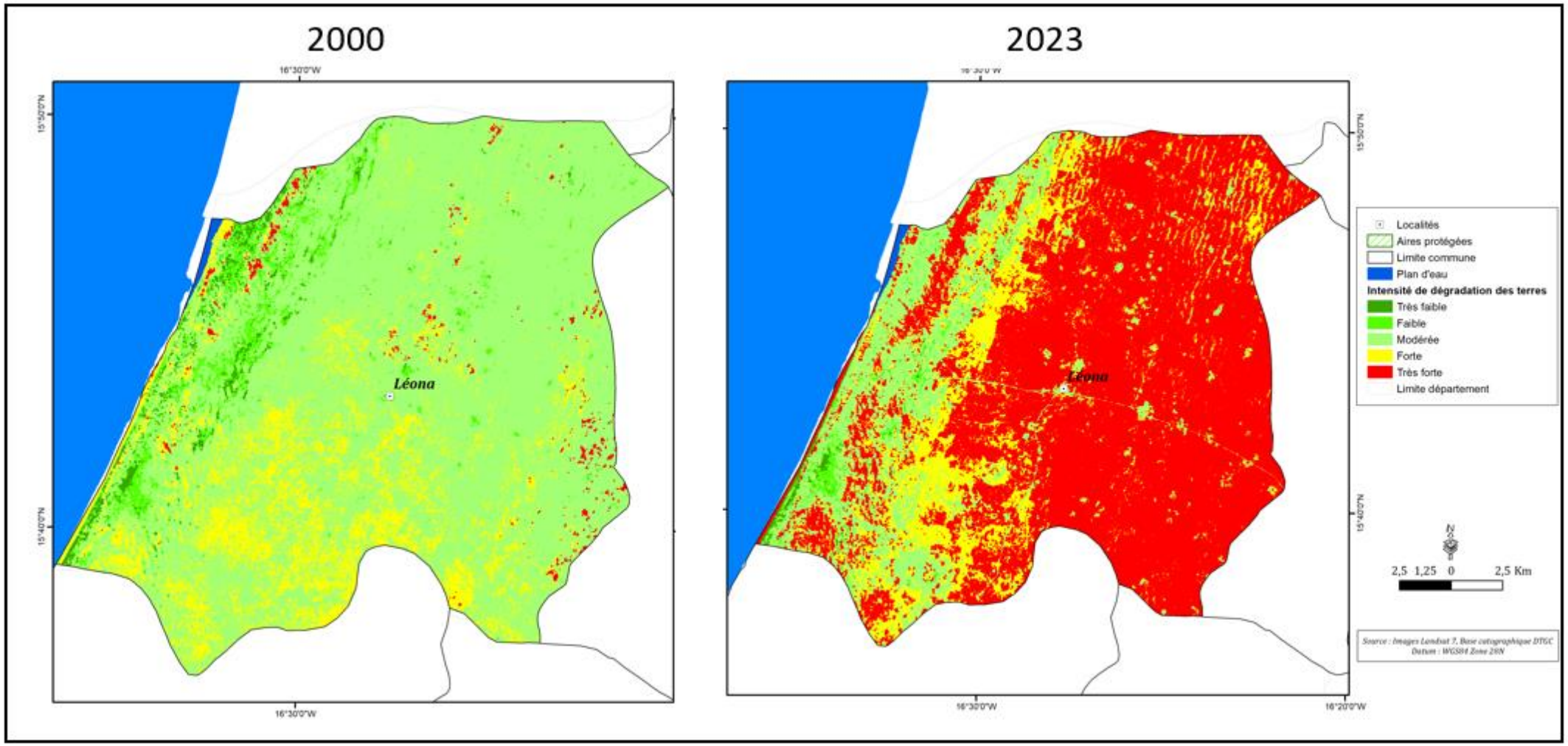
RESULTS



Between 2000 and 2023, soil quality deteriorated overall. Very slightly and slightly degraded soils decreased, while moderately degraded soils fell sharply, from 51.3% to 23.0%.

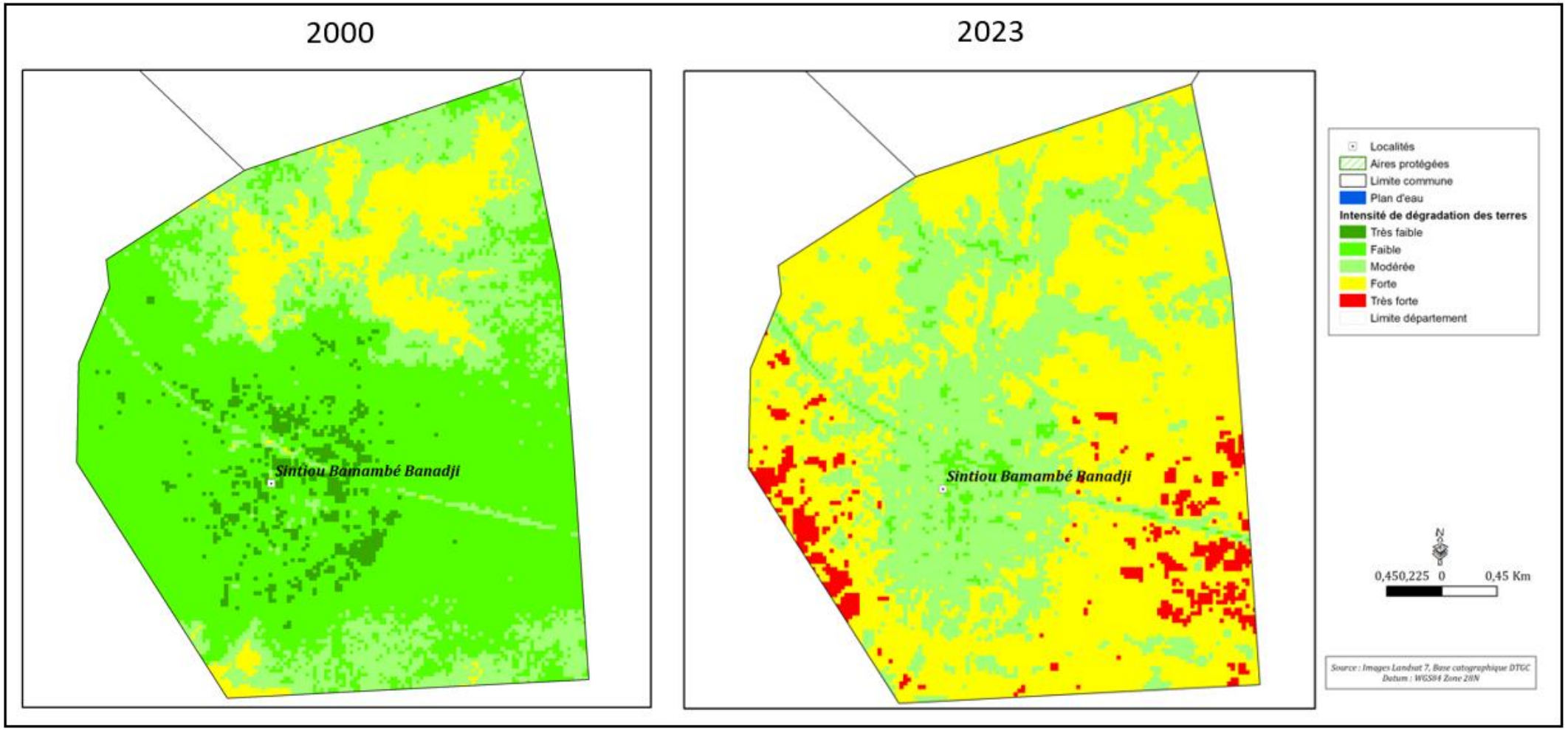
In contrast, the **number of severely and very severely degraded soils has increased**, indicating an intensification of soil degradation.

RESULTS



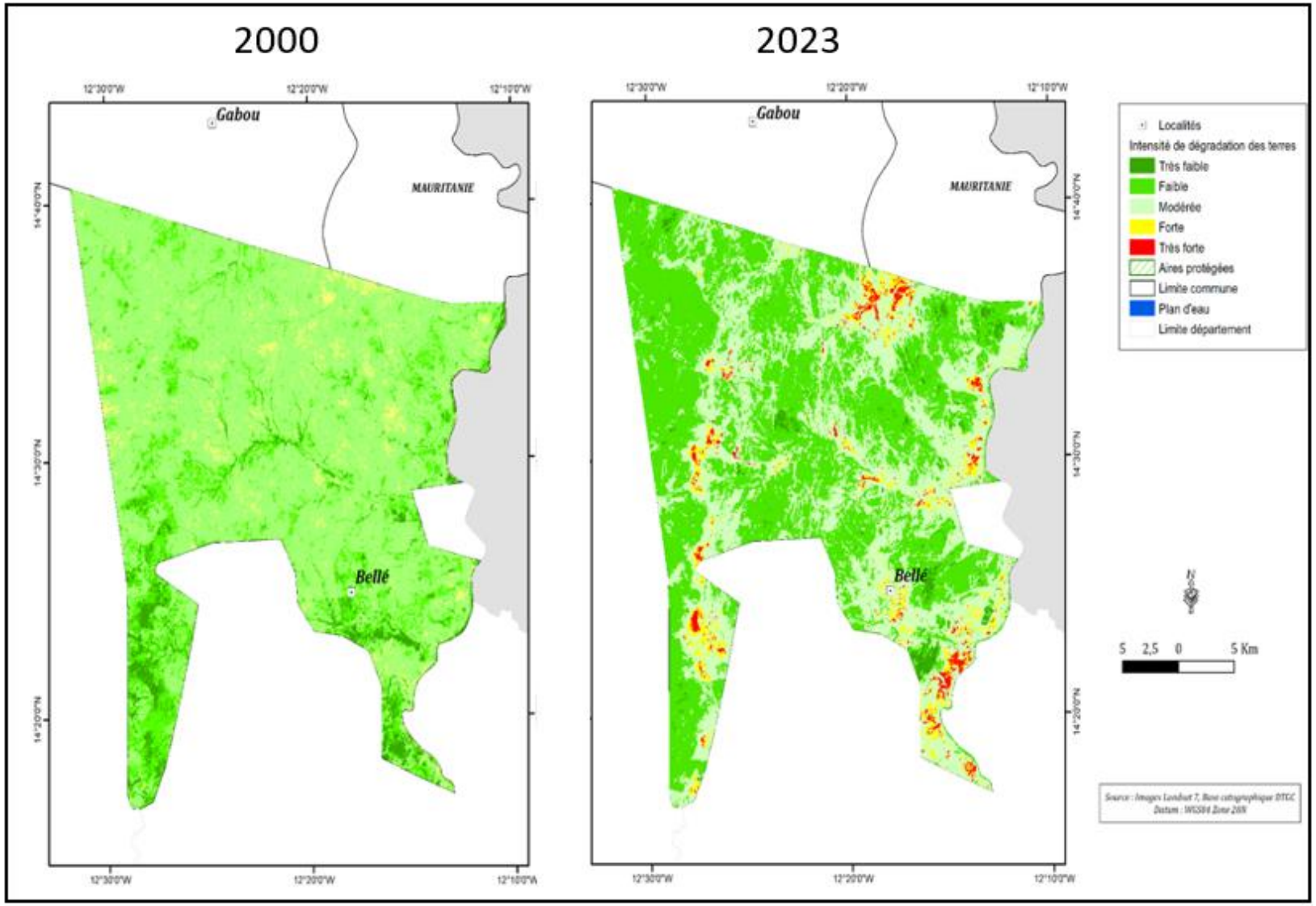
Between 2000 and 2023, the commune of Léona saw its soils become significantly degraded, with a marked increase in moderately, severely and very severely degraded soils, with those in very severe degradation rising from 0.6% to 57.8%.

RESULTS



- Between 2000 and 2023, slightly degraded land declined sharply, while moderately, severely and very severely degraded land increased.
- By 2023, very slightly degraded land has disappeared, and severely degraded land has reached 59.9%.

RESULTS



Between 2000 and 2023, the Belel commune saw an increase in slightly degraded soils, a decrease in moderately and severely degraded soils, and the appearance of very severely degraded soils, representing 0.8% of the land.

CONCLUSION

Deterioration Trends

Marked soil quality decline between 2000-2023.

Sharp reduction in moderately degraded soils observed.

Intensification Pattern

Severely degraded soils increased substantially.

Very severe categories show worrying expansion rates.

Action Required

Climate change context demands urgent intervention.

Political and community measures need immediate implementation.

THANK YOU

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SOURCE
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FRANCE